

AI COMIC STORY GENERATOR

Project Documentation & UI / Functionality Specification



Reference Output: An 8-panel AI-generated comic story with a title, scene illustrations, narration/dialogue, and a final moral.

User Input	FastAPI /generate	AI Story + Images	Comic Preview	PDF Export
------------	----------------------	----------------------	---------------	------------

1. Project Overview

The AI Comic Story Generator is a web application that allows a user to enter a story idea and creative preferences, then generates a short comic story. The application is designed around a simple three-page Jinja2 template flow: story input, comic preview, and export success.

The reference comic contains eight sequential panels. Each panel communicates one part of the story through an illustration and supporting text. The same structure can be used for dynamically generated stories.

2. index.html — Story Input Page

index.html is the main entry point. It presents a clean form where the user defines the basic creative direction for the comic.

Field	Purpose	Example
Story Prompt	Main idea or plot requested by the user.	A girl finds a mysterious robot.
Main Character Name	Name used consistently for the protagonist.	Mira
Setting	Location or world where the story takes place.	A small town
Story Tone	Emotional or narrative style.	Magical / Adventure
Art Style	Preferred visual appearance.	Anime / Cartoon

Form action: The form submits the collected values to the FastAPI **/generate** route using POST.

3. FastAPI Generation Flow

1. Receive the form data from index.html.
2. Build the AI generation request from the prompt, character, setting, tone, and art style.
3. Generate the story as sequential comic panels.
4. Generate or attach the illustration for each panel.
5. Pass the completed comic data to comic_preview.html.

4. comic_preview.html — Comic Preview

This page displays the generated comic before export. The panels are shown sequentially so the user can review the complete storyline.

Panel element	What is displayed
Title	Name of the panel or scene.
Scene illustration	AI-generated image representing the scene.
Scene description	Short description, presented in an italic style.
Caption / narration	Story text that explains the action.

Panel element	What is displayed
Dialogue	Optional speech bubbles or character dialogue.

5. PDF Export & export_success.html

After reviewing the comic, the user can select the PDF export/download action. The backend creates the PDF version of the comic and then shows **export_success.html** with a confirmation message.

Success page purpose: confirm that the comic was exported successfully and provide a simple way for the user to return and create another comic.

6. Responsive CSS Design

- Centered content with a readable maximum width.
- Clean form cards with padding, rounded corners, and subtle shadows.
- Consistent button styling with hover feedback.
- Clear spacing between labels, inputs, buttons, and result panels.
- Responsive layout so the application remains usable on laptop, tablet, and mobile screens.
- Light, inviting visual treatment that supports the creative storytelling theme.

7. Suggested Project Structure

project/

```

■■■ main.py
■■■ templates/
■ ■■■ index.html
■ ■■■ comic_preview.html
■ ■■■ export_success.html
■■■ static/
■ ■■■ style.css
■■■ generated_comics/

```

8. Complete Application Flow

User opens index.html



Enters story prompt, character, setting, tone, and art style



Clicks Generate Comic



FastAPI /generate receives the data



AI generates the comic storyline and panel illustrations



comic_preview.html displays the complete comic



User reviews the panels and exports the comic



PDF is generated/downloaded



export_success.html confirms completion

Final Output: A complete short AI comic, visually similar in structure to the supplied eight-panel reference, with sequential scenes, illustrations, narration/dialogue, and a closing moral.